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Assignee : Amrutha Pasala

Task ID : 624

Design and implement Oracle schemas

Task : As a database engineer, I want to design and implement Oracle schemas for user management and course allocation.

EPIC 620 : Java Backend, SQL, and Logging

Creating tables schema :

```
SQL> CREATE TABLE USERS (
  2    user_id      VARCHAR2(50) PRIMARY KEY,
  3    email        VARCHAR2(100) NOT NULL UNIQUE,
  4    password     VARCHAR2(100) NOT NULL,
  5    role         VARCHAR2(20) NOT NULL CHECK (role IN ('STUDENT', 'FACULTY'))
  6 );
Table created.

SQL> commit;
Commit complete.

SQL> CREATE TABLE STUDENT (
  2    user_id          VARCHAR2(50) PRIMARY KEY,
  3    full_name        VARCHAR2(100) NOT NULL,
  4    date_of_birth    DATE NOT NULL,
  5    gender           VARCHAR2(10),
  6    college          VARCHAR2(100),
  7    qualification    VARCHAR2(100),
  8    course_interested  VARCHAR2(100),
  9    country          VARCHAR2(50),
```

```
 10    state          VARCHAR2(50),
 11    city           VARCHAR2(50),
 12    CONSTRAINT fk_student_user FOREIGN KEY (user_id) REFERENCES USERS(user_id)
 13 );
Table created.
SQL> commit;
Commit complete.
SQL> CREATE TABLE FACULTY (
  2    user_id       VARCHAR2(50) PRIMARY KEY,
  3    full_name     VARCHAR2(100) NOT NULL,
  4    department    VARCHAR2(100),
  5    qualification  VARCHAR2(100),
  6    expertise_area  VARCHAR2(100),
  7    CONSTRAINT fk_faculty_user FOREIGN KEY (user_id) REFERENCES USERS(user_id)
  8 );
Table created.
SQL> commit;
Commit complete.
SQL> CREATE TABLE COURSE (
  2    course_id    VARCHAR2(50) PRIMARY KEY,
  3    title        VARCHAR2(100) NOT NULL,
  4    description   VARCHAR2(500),
  5    duration_weeks NUMBER
  6 );
Table created.
SQL> commit;
Commit complete.
SQL> CREATE TABLE ENROLLMENT (
  2    enrollment_id   VARCHAR2(50) PRIMARY KEY,
  3    student_id     VARCHAR2(50) NOT NULL,
  4    course_id      VARCHAR2(50) NOT NULL,
```

```
  5    enrollment_date  DATE NOT NULL,
  6    status        VARCHAR2(20) NOT NULL CHECK (status IN ('PENDING', 'CONFIRMED', 'REJECTED')),
  7    CONSTRAINT fk_enroll_student FOREIGN KEY (student_id) REFERENCES STUDENT(user_id),
  8    CONSTRAINT fk_enroll_course  FOREIGN KEY (course_id) REFERENCES COURSE(course_id)
  9 );
Table created.
SQL> commit;
Commit complete.
```

Inserting data into tables:

```
SQL> insert into USERS values('1','a@gmail.com','a123','STUDENT');
1 row created.
SQL> commit;
Commit complete.

SQL> insert into FACULTY values('1','JOHN','CSE','PHD','AI');
1 row created.
SQL> commit;
Commit complete.
SQL> insert into COURSE values('101','AI','artificial intelligence',4);
1 row created.
SQL> commit;
Commit complete.
SQL>
SQL> insert into STUDENT values('1',  'abhi',TO_DATE('2004-06-01', 'YYYY-MM-
DD'),'M','SRM','BTECH','CSE','INDIA','AP','MPL');
1 row created.
SQL> commit;
Commit complete.
SQL> insert into ENROLLMENT values('e1','1','101', TO_DATE('2025-06-27', 'YYYY-MM-
DD'),'CONFIRMED');
1 row created.
```

SQL> commit;

Commit complete.

Viewing table schema:

SQL> desc USERS;

Name	Null?	Type
USER_ID	NOT NULL	VARCHAR2(50)
EMAIL	NOT NULL	VARCHAR2(100)
PASSWORD	NOT NULL	VARCHAR2(100)
ROLE	NOT NULL	VARCHAR2(20)

SQL> desc STUDENT;

Name	Null?	Type
USER_ID	NOT NULL	VARCHAR2(50)
FULL_NAME	NOT NULL	VARCHAR2(100)
DATE_OF_BIRTH	NOT NULL	DATE
GENDER		VARCHAR2(10)
COLLEGE		VARCHAR2(100)
QUALIFICATION		VARCHAR2(100)
COURSE_INTERESTED		VARCHAR2(100)
COUNTRY		VARCHAR2(50)
STATE		VARCHAR2(50)
CITY		VARCHAR2(50)

SQL> desc COURSE;

Name	Null?	Type
COURSE_ID	NOT NULL	VARCHAR2(50)
TITLE	NOT NULL	VARCHAR2(100)
DESCRIPTION		VARCHAR2(500)
DURATION_WEEKS		NUMBER

SQL> desc FACULTY;

Name	Null?	Type
USER_ID	NOT NULL	VARCHAR2(50)
FULL_NAME	NOT NULL	VARCHAR2(100)
DEPARTMENT		VARCHAR2(100)
QUALIFICATION		VARCHAR2(100)
EXPERTISE_AREA		VARCHAR2(100)

SQL> desc ENROLLMENT;

Name	Null?	Type
ENROLLMENT_ID	NOT NULL	VARCHAR2(50)
STUDENT_ID	NOT NULL	VARCHAR2(50)
COURSE_ID	NOT NULL	VARCHAR2(50)
ENROLLMENT_DATE	NOT NULL	DATE
STATUS	NOT NULL	VARCHAR2(20)

Viewing tables data :

SQL> select * from USERS;

USER_ID

EMAIL

PASSWORD

ROLE

1

a@gmail.com

a123

STUDENT

SQL> select * from STUDENT;

USER_ID

FULL_NAME

DATE_OF_B GENDER

--- ---

COLLEGE

QUALIFICATION

COURSE_INTERESTED

COUNTRY

STATE

CITY

1

abhi

01-JUN-04 M

SRM

BTECH

CSE

INDIA

AP

MPL

SQL> select * from FACULTY;

USER_ID

FULL_NAME

DEPARTMENT

QUALIFICATION

EXPERTISE_AREA

1

JOHN

CSE

PHD

AI

SQL> select * from COURSE;

COURSE_ID

TITLE

DESCRIPTION

DURATION_WEEKS

101

AI

artificial intelligence

4

SQL> select * from ENROLLMENT;

ENROLLMENT_ID

STUDENT_ID

COURSE_ID ENROLLMEN

--- ---

STATUS

---

e1

1

101 27-JUN-25

CONFIRMED